

SCENT: Robust Spatiotemporal Learning for Continuous Scientific Data via Scalable Conditioned Neural Fields

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Abstract

Spatiotemporal learning is challenging due to the intricate interplay between spatial and temporal dependencies, the high dimensionality of the data, and scalability constraints. These challenges are further amplified in scientific domains, where data is often irregularly distributed (e.g., missing values from sensor failures) and high-volume (e.g., high-fidelity simulations), posing additional computational and modeling difficulties. In this paper, we present SCENT, a novel framework for scalable and continuity-informed spatiotemporal representation learning. SCENT unifies interpolation, reconstruction, and forecasting within a single architecture. Built on a transformer-based encoder-processor-decoder backbone, SCENT introduces learnable queries to enhance generalization and a query-wise cross-attention mechanism to effectively capture multi-scale dependencies. To ensure scalability in both data size and model complexity, we incorporate a sparse attention mechanism, enabling flexible output representations and efficient evaluation at arbitrary resolutions. We validate SCENT through extensive simulations and real-world experiments, demonstrating state-of-the-art performance across multiple challenging tasks while achieving superior scalability.

1. Introduction

Spatiotemporal learning focuses on modeling and interpreting data that exhibit variations in both space and time. This approach is crucial for analyzing intricate real-world phenomena where spatial structures are inextricably linked with temporal dynamics, including applications such as climate modeling (Reichstein et al., 2019), traffic forecasting (Li et al., 2018), medical imaging (Litjens et al., 2017), and video analysis (Wang et al., 2020). Achieving accurate

spatiotemporal learning, however, presents significant challenges due to the presence of complex spatial-temporal dependencies, including spatial heterogeneity and temporal non-stationarity, compounded by the high dimensionality of the data. These challenges are further amplified in scientific and engineering domains, where datasets are frequently characterized by irregular distributions (e.g., arising from sensor malfunctions) and large volumes (e.g., generated by high-fidelity simulations), introducing further complexities in both computation and modeling.

To address the aforementioned challenges, significant research efforts have been dedicated to developing solutions from both traditional signal processing and, more recently, machine learning perspectives (a detailed literature review is provided in the Appendix A). Among the emerging machine learning approaches, Implicit Neural Representations (INRs) have garnered increasing attention due to their inherent flexibility (Sitzmann et al., 2020b; Mildenhall et al., 2020). INRs parameterize data as a continuous function, mapping coordinates to signal values (e.g., $(x, y) \rightarrow (r, g, b)$ for images), using a neural network. Despite their potential, the adoption of INRs in scientific domains has been limited. This can primarily be attributed to two key challenges: scalability and generalizability.

To bridge this methodological gap, we propose **SCENT** (Scalable Conditioned Neural Field for SpatioTemporal Learning), a novel framework designed to address the limitations of existing INR approaches. SCENT leverages a Transformer-based encoder-processor-decoder architecture to efficiently process large-volume spatiotemporal data, demonstrating strong scalability with respect to both dataset size and model parameter count. Furthermore, SCENT incorporates trainable query mechanisms to enhance generalizability, circumventing the computational overhead associated with existing strategies such as latent optimization (Dupont et al., 2022) or meta-learning (Chen & Wang, 2022).

Overall, the major contributions of our work include:

- We offer a unified framework for spatiotemporal learning, capable of performing joint interpolation, reconstruction, and forecasting.
- We empirically demonstrate that SCENT is scalable

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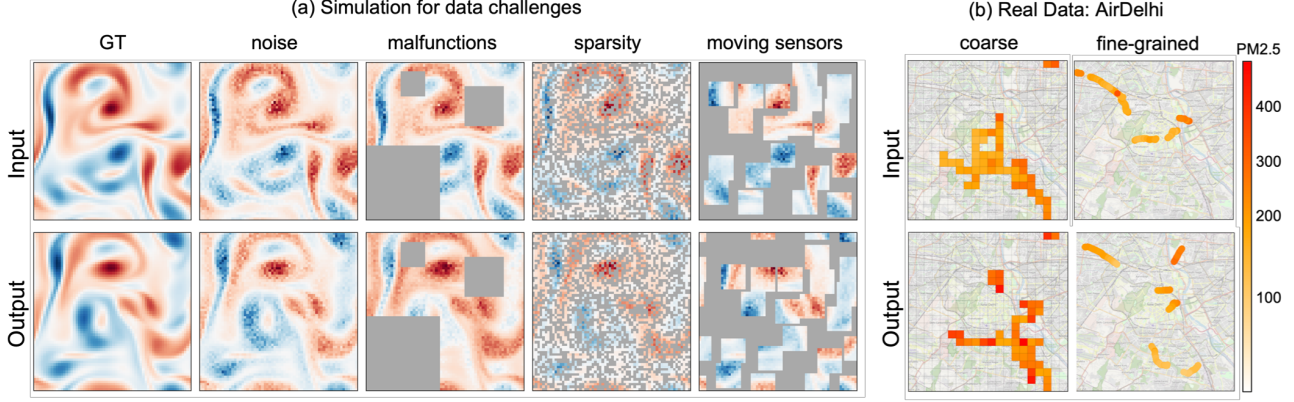


Figure 1. **Challenging data scenarios motivating this study.** (a) Learning continuous ground truth signal (GT) given noisy, malfunctioning, sparse, or moving sensors is a daunting challenge. However, these challenges are common in scientific data, including AirDelhi (Chauhan et al., 2024) data which measure the particulate matter levels (PM2.5) from moving vehicles.

with larger models and datasets.

- We conduct extensive experiments to evaluate the efficacy of the proposed SCENT framework, demonstrating its superiority over state-of-the-art (SOTA) methods in a range of spatiotemporal learning tasks.

2. Problem Setting

We consider a spatiotemporal discrete observation denoted as $u_{\mathbf{x}}^t$, where $\mathbf{x} = (x, y, z) \in \mathbb{R}^3$ represents the spatial coordinates, and $t \in \mathbb{R}$ denotes the sampled time. Given a set of N_i input observations measured at time t_i , we define:

$$U^{t_i} = \{u_{\mathbf{x}_1}^{t_i}, u_{\mathbf{x}_2}^{t_i}, \dots, u_{\mathbf{x}_{N_i}}^{t_i}\}.$$

Our objective is to develop a model F capable of learning the underlying continuous spatiotemporal function from these discrete N_i observations, enabling it to predict M target outputs at a different time t_o :

$$\hat{U}^{t_o} = F(U^{t_i}),$$

where the target outputs are given by:

$$U^{t_o} = \{u_{\mathbf{x}'_1}^{t_o}, u_{\mathbf{x}'_2}^{t_o}, \dots, u_{\mathbf{x}'_{N_o}}^{t_o}\}.$$

The specific task performed by (F) is determined by the relationship between the input and output spatiotemporal coordinates:

- **Reconstruction.** If $(\{\mathbf{x}_j\}_{j=1}^{N_i}, t_i) = (\{\mathbf{x}'_k\}_{k=1}^{N_o}, t_o)$, the model is tasked with reconstructing the same observations it was provided.
- **Spatiotemporal Interpolation.** If $(\{\mathbf{x}_j\}_{j=1}^{N_i}, t_i) \neq (\{\mathbf{x}'_k\}_{k=1}^{N_o}, t_o)$, the model estimates the function at novel spatiotemporal locations.

- **Forecasting.** A special case of interpolation where $t_o > t_i$, requiring the model to predict future values.

Our goal is to develop a model (F) capable of jointly performing reconstruction, interpolation, and forecasting of complex scientific data given arbitrary spatiotemporal coordinates (x, y, z, t) .

3. SCENT

3.1. Encoder-Processor-Decoder Framework

The encoder-processor-decoder architecture is proposed as a general framework that scales linearly with input and output sizes, overcoming the quadratic scaling limitations of transformers with input sequence length (Jaegle et al., 2022). Prior works have leveraged this architecture to learn conditioned neural fields (CNFs) for solving partial differential equations (Lee & Oh, 2024; Serrano et al., 2024; 2023; Li et al., 2023), where the architecture is particularly beneficial, as these models flatten the input and treat each sample point as an individual token. Additionally, this architecture facilitates learning continuity-informed representations using inducing point learning within cross-attention layers (Jaegle et al., 2022; Lee & Oh, 2024). Building on these advancements, we evaluate the models' potentials in complex real-world scientific problems, often characterized by intricate noise patterns and irregular sensor distributions. In the remaining subsections, we describe notable developments on top of the existing architecture, as displayed by red-colored objects in Fig. 2(a).

3.2. Time-Targeted Spatial Encoder

The encoder processes an input data $U^{t_i} = \{u_{\mathbf{x}_1}^{t_i}, u_{\mathbf{x}_2}^{t_i}, \dots, u_{\mathbf{x}_{N_i}}^{t_i}\}$, representing N_i samples from the space coordinate $\{\mathbf{x}_j\}_{j=1}^{N_i}$ at a given time t_i . Here,

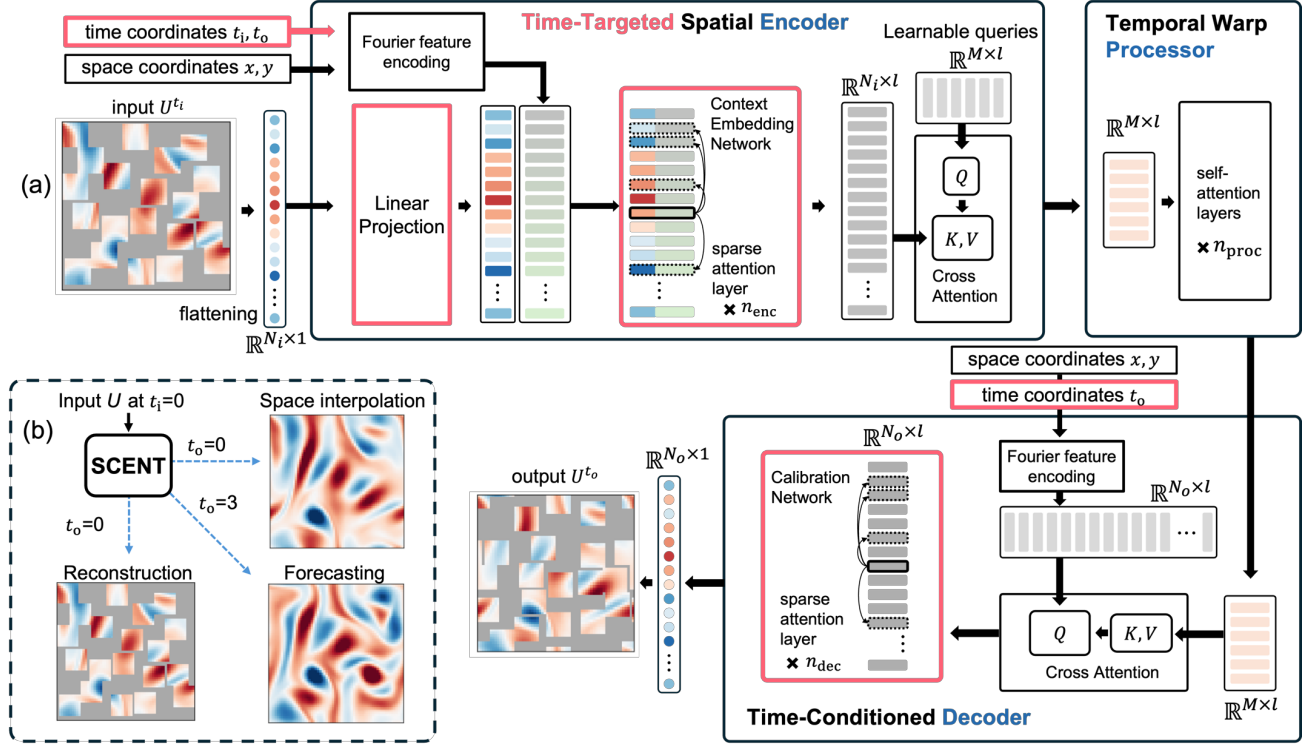


Figure 2. **SCENT overview.** (a) Detailed architecture of SCENT is illustrated. Our unique contributions are drawn with red boxes. Specifically, we introduce time coordinates to both encoder and decoder for learning continuous time representations. Also, we introduce Context Embedding Network and Calibration Network for improved spatial encoding and decoding, respectively. n_{enc} , n_{proc} , and n_{dec} denote the number of layers in encoder, processor, and decoder respectively. (b) At the inference stage, a single SCENT model is capable to jointly perform reconstruction, spatiotemporal interpolation, and forecasting.

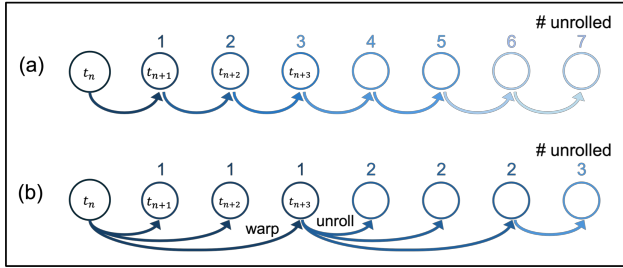


Figure 3. **Warp-unrolling forecasting (WUF).** (a) Conventional forecasting includes single-step unrolling which accumulates error. A dimming blue color is used to represent the increasing error. (b) WUF helps mitigate error accumulations caused by extensive unrolling steps.

$\{\mathbf{x}_j\}_{j=1}^{N_i}$ can be structured on a grid or an irregular mesh. Using a cross-attention mechanism, the encoder transforms U^{t_i} into a fixed-size set of tokens $Z_M^{t_i} = \{z_1, z_2, \dots, z_M\}$, where M is the number of latent trainable query tokens. The encoder E_θ is then expressed as:

$$E_\theta : \left(U^{t_i}, \{\mathbf{x}_j\}_{j=1}^{N_i}, t_i, t_o \right) \rightarrow Z_M^{t_i},$$

where t_i is the input time, and t_o is the targeted output time. Including t_o enhances attention to relevant information, leading to improved performance (Table 3).

Specifically, we encode the spatiotemporal coordinates $\{\mathbf{x}_j\}_{j=1}^{N_i}, t_i, t_o$ using Fourier features (Tancik et al., 2020). Fourier features map input coordinates to a higher-dimensional space using sinusoidal functions of varying frequencies, enabling models to capture fine-grained and periodic variations in the data. Meanwhile, the input samples U^{t_i} are separately embedded using a linear projection layer with parameters frozen. This effectively increases the dimensionality of function value representation U^{t_i} , empirically enhancing performance. Both Fourier features and encoded samples are concatenated, and sent through Context Embedding Network (CEN) which consists of sparse self-attention layers to enrich context encoded in individual tokens. Within CEN, each token attends to randomly subsampled S tokens where $S \ll N_i$. As the underlying data is in continuous field, sparse attention is an efficient way to encode global contexts to individual tokens. The final latent representation is then summarized through a cross attention against the learning queries which consist of M tokens, where $M \ll N_i$. We justify the benefits of the linear

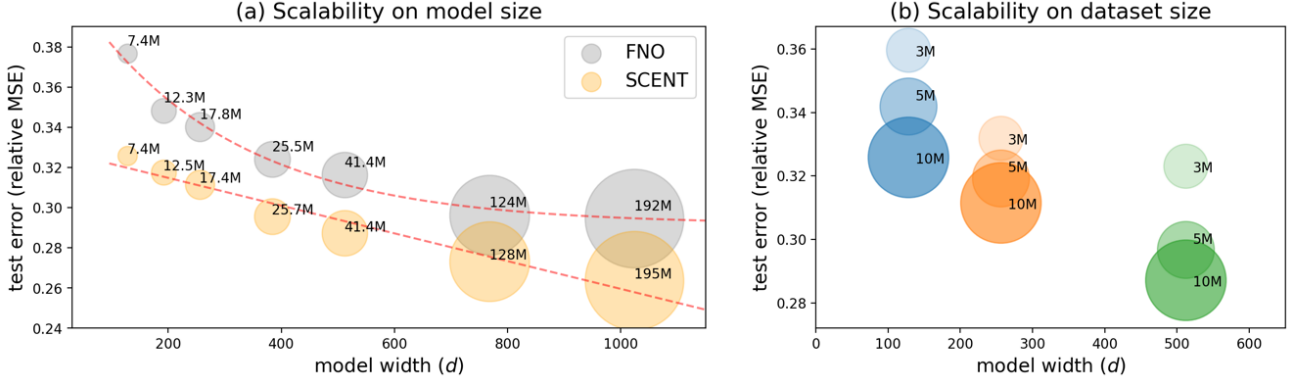


Figure 4. **Scalability evaluations.** (a) Texts next to each circle are the number of model parameters, and circle size is also proportional to it. Red dotted lines are scalability trends derived with exponential functions for comparisons. (b) Colors indicate training runs with identical model sizes. Texts next to each circle are the number of training instances, and the circle size is also proportional to it.

projection and CEN by the ablation study (Table 3).

3.3. Temporal Warp Processor

We define the Time Warp Processor (TWP) for learning continuous temporal dynamics, denoted as $P_\theta : Z_M^{t_i} \rightarrow Z_M^{t_o}$, where $t_o = t_i + \Delta t$, $\Delta t \in [0, t_h]$ and t_h represents a hyperparameter for the maximal time horizon used during training. Depending on Δt , TWP can either perform input reconstruction ($\Delta t = 0$) or forecasting ($\Delta t > 0$), allowing joint reconstruction and forecasting with a single model. This flexibility also enables the use of input-output pairs sampled at *non-integer* time intervals, which is particularly useful for spatiotemporal data with time-varying sampling rates (Chauhan et al., 2024; Nie et al., 2024).

Novel Unrolling Strategy. TWP can be leveraged to minimize accumulating prediction errors during long-horizon forecasting, which empirically leads to improved performance over baselines (Section 4.5). Previous models have been reported to struggle with long-term forecasting due to rapidly accumulating errors during inference and unrolling (Serrano et al., 2024). While we adhere to the standard practice of training models for next-state prediction, we do not perform one-step unrolling for the entire forecasting time steps. Instead, we utilize time warping, advancing directly to the time horizon t_h once $t_o - t_i > t_h$, using it as a reference state for predicting subsequent time steps (Fig. 3). This ensures that at any given time state, only a minimal number of prediction steps are required, thereby reducing the potential accumulation of errors. We denote this strategy as *warp-unrolling forecasting (WUF)* and use this for benchmark datasets (Table 2).

3.4. Time-Conditioned Decoder

The decoder $D_\theta : Z_M^{t_o} \rightarrow U^{t_o}$ utilizes the latent processed tokens $Z(t_o)$ to approximate the function values at t_o . To

this end, we apply Fourier features encoding to $x \text{ and } t_o$, and use the resulting queries for cross-attention against $Z(t_o)$. Then we apply n_{dec} sparse self-attention layers (Calibration Network, abbreviated as CN) to calibrate spatiotemporal decoding. During training U^{t_o} includes N_o available samples at t_o . During inference, however, $Z(t_o)$ may be evaluated on arbitrary points in $\mathbf{x}$.

4. Experiments

We conduct evaluations using a diverse set of baseline models, encompassing state-of-the-art regular-grid methods such as FNO (Li et al., 2020), adaptable transformer architectures represented by OFormer (Li et al., 2023), as well as neural field-based approaches like DINO (Yin et al., 2023), CORAL (Serrano et al., 2023) and AROMA (Serrano et al., 2024). All training and evaluations are conducted using mean squared error (MSE), relative MSE (Rel-MSE), and root MSE (RMSE). RMSE is specifically used to measure PM2.5 levels for AirDelhi datasets (Section 4.1.3). Rel-MSE is defined as:

$$\text{Rel-MSE} = \frac{\sum_{j=1}^N (U_j^{t_o} - \hat{U}_j^{t_o})^2}{\sum_{j=1}^N (U_j^{t_o})^2}. \quad (1)$$

Most experiments were performed on a single NVIDIA H100 80GB HBM3 GPU. The largest model variant used in the scalability evaluation (Fig. 4) required distributed training across eight of them. Details of the algorithm can be found in Appendix B, while information on the dataset and training procedures is provided in Appendices C through G.

4.1. Datasets

4.1.1. BENCHMARK NAVIER-STOKES DATASETS

We use three benchmark Navier-Stokes datasets (Li et al., 2020), each corresponding to different viscosity coefficients.

These are designed to model the dynamics of a viscous and incompressible fluid governed by the 2D Navier-Stokes equation in vorticity form on the unit torus. The Navier-Stokes 1×10^{-3} (NS-3) dataset (Yin et al., 2023; Serrano et al., 2023), with a viscosity coefficient $\nu = 1 \times 10^{-3}$, includes 256 training trajectories and 32 testing trajectories. NS-3 models relatively slower fluid dynamics with a time horizon of first 20 steps beyond the initial condition. The Navier-Stokes 1×10^{-4} (NS-4) dataset, with a viscosity coefficient $\nu = 1 \times 10^{-4}$, is a more turbulent variant. Lastly, the Navier-Stokes 1×10^{-5} (NS-5) dataset, with the lowest viscosity coefficient $\nu = 1 \times 10^{-5}$, represents the most turbulent fluid dynamics. Both NS-4 and NS-5 contain 1000 training and 200 testing trajectories and a maximum time steps of 30 and 20 steps, respectively. For all three datasets, we use the vorticity at $t_0 = 10$ as the initial condition for testing. These datasets provide a range of viscosity conditions, making them suitable for studying fluid dynamics across different turbulence levels.

4.1.2. SIMULATED LARGE-SCALE COMPLEX DATASETS

We introduce a new class of datasets to specifically simulate complex yet common real-world data scenarios as visualized in Fig. 1(a). This simulated dataset, derived from the Navier-Stokes equations, is designed to be significantly larger than benchmark datasets (Section 4.1.1), incorporating real-world challenges such as sensor noise, missing values, and dynamically moving or reconstructed sensor locations.

Concretely, we introduce five datasets. While each variant represents different data challenges, they consist of the same underlying ground truth. • **Ground truth (S1)**: We utilize the Navier-Stokes equation with viscosity and boundary conditions to simulate highly complex and fast diffusion dynamics (Appendix E). This setup ensures that our simulations capture the intricate interactions of fluid motion, enabling a more realistic representation of challenging spatiotemporal processes. • **Noisy sensors (S2)**: multiplicative noise is sampled (Nartasilpa et al., 2016) and applied to S1 following: $u'(\mathbf{x}, t) = u(\mathbf{x}, t) \cdot \eta(\mathbf{x}, t)$, where $\eta(\mathbf{x}, t) \sim \mathcal{N}(\mu, \sigma)$, u is an input sample, η is sample-level noise, and $\mathbf{x} = (x, y, z)$ and t denote the spatial and temporal coordinates respectively. We use $\mu = 1$ and $\sigma = 0.2$. • **Sensor malfunctions (S3)**: scientific sensors are often temporarily unavailable or malfunctioning (Nathaniel et al., 2023). We empty out three square areas with different sizes to simulate lost sensors. • **Randomly sparse sensors (S4)**: sensor locations may be randomly placed, such as for remote sensing (Myneni et al., 2001) in climate science. We randomly mask out 50% of the data to simulate the sparsity. • **Dynamically moving sensors (S5)**: A more challenging scenario arises when sampling locations are sparse and dynamically shifting (Chauhan et al., 2024; Nie et al., 2024). To simulate this,

we randomly select twenty non-overlapping 10×10 square regions as input locations. The output regions are then translated by (h, v) , where $h, v \in [-10, 10]$ denote horizontal and vertical shifts. Regions crossing image boundaries are mirrored to maintain continuity.

4.1.3. REAL DATA: AIRDELHI

The AirDelhi (Chauhan et al., 2024) dataset offers a comprehensive collection of fine-grained spatiotemporal particulate matter (PM) measurements from Delhi, India. To address the limitations of static sensor networks, the researchers mounted lower-cost PM2.5 sensors on public buses throughout the Delhi-NCR region. The dataset includes PM2.5 measurements across various locations and times, with data collected at a granularity of 20 samples per minute. The dynamic nature of the data, with sensors moving along pre-defined bus routes, introduces challenges such as sparse and temporally varying measurements. This necessitates the development of models capable of handling sparse and dynamically moving sensors. Here we use three data variants for model performance comparisons.

• **AirDelhi Benchmark (AD-B)**: This dataset was originally introduced as a benchmark for evaluation. Concretely, this data is collected between November 12, 2020, and January 30, 2021. In this data, initial days are omitted due to limited sample data and fewer instruments on buses. Also excluded are nightly data between 10:00 PM and 5:30 AM, as buses remain stationary at bus depots during this period. The geographical region is divided into 1 km^2 spatial grids, which are further segmented into spatiotemporal cells with a 30-minute time interval. The average of all samples within each spatiotemporal cell is calculated to determine representative PM values. We use the ‘AB’ and ‘CP’ sets as training and test datasets, respectively. This dataset features differing numbers of samples for input (N_i) and output (N_o), which demands a high degree of model flexibility. Notably, SCENT stands out as the only model capable of naturally handling a variable N_i and N_o . • **AirDelhi Temporal (AD-T)**: this dataset still uses the 1 km^2 grid, but increases time resolution from 30 minutes to a finer 1 minute, and averaging all samples within each spatiotemporal cell for representative PM values. This increases available data instances but makes data spatially more sparse and thus challenging to predict. Unlike the original dataset, where the train-test split was based on specific days, we randomly shuffle all the data before splitting it into training and testing sets. This approach ensures there is no distribution shift between the training and test data. • **AirDelhi Fine-grained (AD-F)**: this dataset features a 0.02 km^2 spatial grid with a 1-minute temporal resolution. The high granularity of the spatial resolution makes it nearly continuous, enabling the evaluation of models in learning continuous representations effectively.

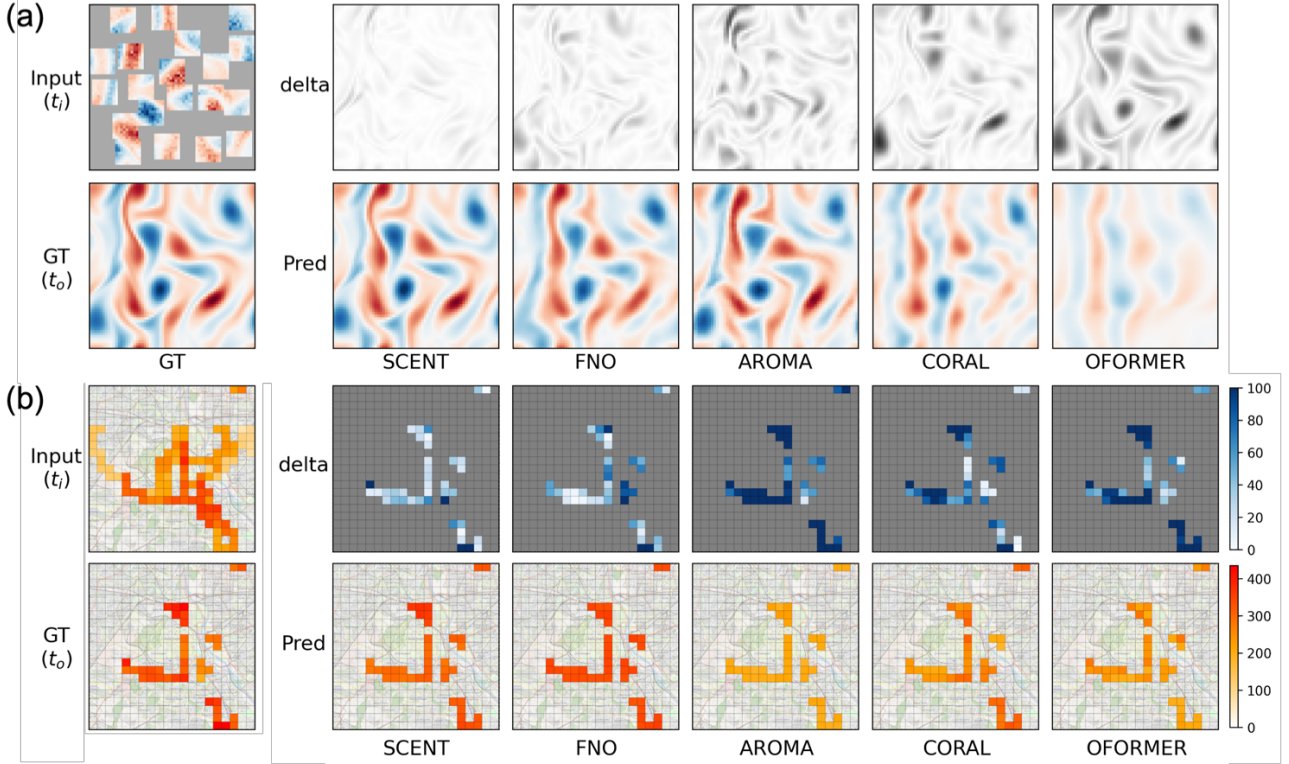


Figure 5. **Qualitative comparisons for forecasting.** (a) Using partial input from the S5 dataset, each pretrained model is assessed on the full mesh (GT), performing both joint forecasting and spatial interpolation. (b) Models forecast PM2.5 on the spatiotemporal locations where measurements are available (GT). ‘delta’ (upper rows) represents the absolute difference between each prediction and GT.

4.2. Robustness to Data Challenges

A robust model should reconstruct continuous fields from sparse, noisy data while capturing temporal dynamics. To evaluate this capability, we assess forecasting performance across five challenging datasets (S1-S5, Fig. 1(a)) using four representative baseline models. All models are trained with supervision on next-state prediction. S1 represents an ideal setting with no noise and unlimited sensor availability, while S2 introduces noise with full sensor coverage to measure its impact. S3 evaluates performance when a portion of sensors are unavailable, S4 considers a scenario with sparsely available but spatially well-distributed static sensors, and S5 examines the effect of time-varying sampling locations. Since FNO lacks mesh independence, we modify the original architecture to accommodate our sparse and irregular dataset. Sparse data variants are zero-padded to align with a regular grid representation for input processing. To prevent the model from trivially predicting zero values, a data mask is applied to the output during loss computation. The *time horizon* (t_h) is set to 3 for training. This implies that during training, the time step (Δt) is randomly selected from $[0, 3]$, allowing the model to learn time-continuity given varying temporal intervals as well as reconstruction ($\Delta t = 1$). Each training trajectory consists of 24 time steps, and a total of 100k trajectories are used. Training is conducted using Rel-

MSE as the supervision loss, a cosine learning rate scheduler over 50k iterations, and a batch size of 256. During validation, Δt is fixed to 1.

Results. Table 1 presents forecasting performance comparisons. SCENT consistently outperforms all baseline models on the simulated datasets, showcasing its resilience in challenging environments and its ability to effectively capture spatiotemporal patterns. As expected, performance generally degrades in more challenging environments, reflecting the increased difficulty of learning from sparse or noisy data. Surprisingly, FNO performs competitively despite its lack of mesh independence. Given FNO’s inherent bias toward learning low-frequency components, it may exhibit robustness to noise and challenging conditions, allowing it to maintain relatively strong performance even in less favorable settings.

4.3. Scalability Analysis

Scientific data volumes are rapidly increasing (Yu et al., 2023; Kitamoto et al., 2023; Kaltenborn et al., 2023), reaching the exabyte scale in cases such as ATLAS (Peters & Janyst, 2011). To assess scalability, we analyze model and dataset size variations, examining both model width (i.e., latent dimension) and depth (i.e., number of layers). To our

Table 1. Forecasting performance comparisons on simulated and real datasets. Rel-MSE and RMSE are used for simulated and real data, respectively.

MODEL	SIMULATED CHALLENGING ENVIRONMENTS					REAL DATA (AIRDELHI)		
	S1	S2	S3	S4	S5	AD-B	AD-T	AD-F
FNO	4.27×10^{-2}	2.10×10^{-1}	2.81×10^{-1}	2.35×10^{-1}	3.77×10^{-1}	48.79	55.04	55.66
OFORMER	1.63×10^{-1}	2.57×10^{-1}	3.30×10^{-1}	3.00×10^{-1}	6.17×10^{-1}	70.62	57.00	54.58
CORAL	4.12×10^{-1}	4.83×10^{-1}	4.69×10^{-1}	4.89×10^{-1}	9.06×10^{-1}	60.51	55.26	48.41
AROMA	1.29×10^{-1}	2.38×10^{-1}	2.83×10^{-1}	6.67×10^{-1}	5.25×10^{-1}	40.78	63.06	<u>47.49</u>
SCENT	2.51×10^{-2}	2.08×10^{-1}	2.70×10^{-1}	2.28×10^{-1}	3.26×10^{-1}	<u>44.20</u>	53.24	45.35

Table 2. Long-term forecasting performances on Navier-Stokes benchmark datasets. MSE is used for NS-3 (for fair comparisons against reported prior arts), while Rel-MSE is used for NS-4 and NS-5.

MODEL	BENCHMARK DATASETS		
	NS-3	NS-4	NS-5
FNO	1.55×10^{-4}	1.53×10^{-1}	1.24×10^{-1}
DINO	2.51×10^{-2}	7.25×10^{-1}	3.72×10^{-1}
CORAL	5.76×10^{-4}	3.77×10^{-1}	3.11×10^{-1}
OFORMER	7.76×10^{-3}	1.36×10^{-1}	2.40×10^{-1}
GNOT	3.21×10^{-4}	1.85×10^{-1}	1.65×10^{-1}
AROMA	1.32×10^{-4}	1.05×10^{-1}	1.24×10^{-1}
SCENT	7.78×10^{-5}	1.03×10^{-1}	1.17×10^{-1}

knowledge, this is the first systematic scalability study of continuity-informed architectures (e.g., INRs, CNFs), and we compare against FNO by varying model width to assess large-scale adaptability. After constructing SCENT models of varying sizes (Appendix F), we design corresponding FNO models with approximately matching parameter counts for fair comparison.

Results. Fig. 4(a) illustrates the results, showing that both FNO and SCENT exhibit decreasing Rel-MSE as model size increases. However, SCENT consistently outperforms FNO across all model sizes and demonstrates better scalability, following a linear trend, whereas FNO shows a clear convergence pattern. Additionally, Fig. 4(b) indicates that larger datasets lead to improved forecasting performance, indicating that the model effectively leverages additional samples to capture complex underlying patterns from sparse sensor data.

4.4. Learning Spatiotemporal Continuity

Our primary objective is to represent a spatiotemporal field conditioned on a sparse input, as outlined in Section 2. An ideal model should, given an input at t_i , be capable of both reconstruction and spatial interpolation at t_o , as well as forecasting at any continuous time within the given time horizon. To evaluate this capability, we assess and compare model performance, with a particular focus on learning

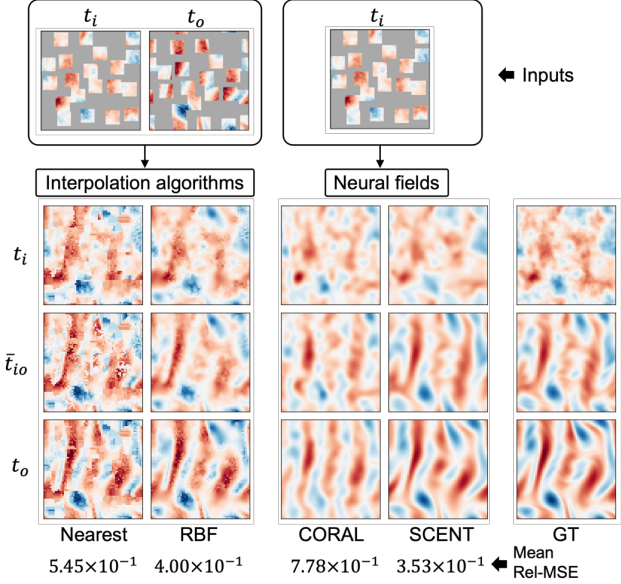


Figure 6. **Joint reconstruction, interpolation, and forecasting.** Given dataset S5 inputs shown on top, neural fields are tested for reconstruction/interpolation (t_i) and forecasting at continuous time ($\bar{t}_{io}, t_o$). For comparison purposes, interpolation results from nearest-neighborhood and RBF algorithms are shown on the left. ‘Mean Rel-MSE’ denotes average performance at $\bar{t}_{io}$ evaluated with the validation dataset.

time-continuous representations. Specifically, given an S5 data input U^{t_i} , we infer the data field at three time points, $t_i, \bar{t}_{io}, t_o$, where $\bar{t}_{io} = (t_i + t_o)/2$ denotes the midpoint time. Reconstruction performances at $\bar{t}_{io}$ are examined against ground truth. We downsample each trajectory by a factor of two, reserving the rest for evaluating time continuity in learning. We compare the results against CORAL, which also learns time-continuous conditioned neural fields (Table 4). Additionally, we evaluate our model against deterministic interpolation methods, namely nearest-neighbor and radial basis function (RBF) interpolation. Unlike our approach, these methods require data at both t_i and t_o to interpolate the spatiotemporal full mesh effectively.

Results. Fig. 6 presents the reconstructed results for a given input. Notably, SCENT accurately interpolates and re-

Table 3. Ablation study for SCENT on dataset S5. ✓ indicates activated modules. Abbreviations: CEN=Context Encoding Network; CN=Calibration Network; Proj=linear projection; TT=Time-Targeted: whether to provide t_{out} for the TTSE.

CEN	CN	PROJ	TT	REL-MSE	CONTRAST
✓	✓	✓	✓	3.26×10^{-1}	-
✗	✓	✓	✓	4.02×10^{-1}	+23.3%
✓	✗	✓	✓	3.64×10^{-1}	+11.0%
✓	✓	✗	✓	3.42×10^{-1}	+4.9%
✓	✓	✓	✗	3.40×10^{-1}	+4.29%
✗	✗	✓	✓	4.61×10^{-1}	+41.4%
✗	✗	✗	✓	4.82×10^{-1}	+47.8%
✗	✗	✗	✗	5.47×10^{-1}	+67.8%

constructs all three time points, closely matching the ground truth (GT). While CORAL supports time interpolation, it inherits suboptimal performance from training with the complex S5 dataset (Table 1), resulting in an inaccurate reconstruction of fields across different time points. Interpolation using radial basis functions (RBF) yields reasonable results; however, it is important to note that the inputs for interpolation methods and neural fields differ. Neural fields operate at a disadvantage as they are provided only with the initial time state. To further evaluate time continuity, we report the mean reconstruction error against the GT at $\bar{t}_{io}$ on the full test dataset, with the quantitative results aligning well with the visual assessments.

4.5. Benchmark Performance and Model Ablations

In this section, we highlight the key innovations in our algorithm, compare it against popular benchmark datasets, and present ablation studies on the SCENT architecture. To evaluate SCENT’s ability to perform extended time forecasting, we test it on NS-3,4,5 (Section 4.1.1) using an unrolling approach. Specifically, all models are trained with supervision on next-state prediction. At test time, we unroll the dynamics following the WUF framework, as illustrated in Section 3.3. Additionally, we conduct an ablation study on S5, systematically removing our key architectural components — CEN, CN, Proj, and TT (Fig. 2) — to assess their individual contributions to performance.

Results. Table 2 presents the results of long-term forecasting performance across the benchmark datasets. SCENT outperforms all baseline models across all datasets, which we attribute to WUF, fundamentally enabled by the time-continuity learned by the model. This advantage is particularly evident in the NS-3 dataset, where the fluid dynamics are relatively slower, hence error accumulation during next-state unrolling is more pronounced. On NS-4 and NS-5, SCENT and AROMA achieve comparable performances. Our ablation study in Table 3 highlights the contributions of individual architectural modifications introduced in SCENT.

The performance gap relative to the best-performing model, referred to as *contrast*, demonstrates that all four components play a crucial role in the model’s effectiveness. Notably, performance deteriorates significantly when two or more modules are deactivated, with Rel-MSE increasing by 67.8% when all modules are not used.

4.6. Forecasting Performance on AirDelhi

We evaluate whether the superior performance observed in previous experiments extends to the more complex AirDelhi dataset. Similar to dataset S5, AirDelhi features sparse sensors with locations that vary across time, posing a significant challenge for spatiotemporal learning. Strong performance on this dataset would indicate that the model effectively infers the PM2.5 distribution from sparse observations and accurately predicts the future diffusion of particulate matter. While we use Mean Squared Error (MSE) as the training loss, we adopt Root Mean Squared Error (RMSE) for evaluation to better capture the physical significance of the model’s performance.

Results. Table 1 compares forecasting performance across AirDelhi datasets. On AD-B, where the goal is to predict PM2.5 values at sparse locations given three days of observations, SCENT ranks second after AROMA, likely due to AROMA’s diffusion backbone effectively filtering noise. However, the qualitative results in Fig. 5(b) highlight SCENT’s effectiveness in capturing and expressing PM2.5 levels. All of the five models outperform previously reported performances on AD-B (Appendix K). For larger and finer datasets (AD-T and AD-F), SCENT outperforms all baselines, demonstrating its strength in learning continuous representations and forecasting. Additional results (Appendix Fig. J) indicates that SCENT better captures the PM2.5 distribution.

5. Conclusion

SCENT (Scalable Conditioned Neural Field for Spatiotemporal Learning) addresses the challenge of reconstructing and forecasting spatiotemporal fields from sparse and noisy data. Through extensive evaluations, we demonstrate SCENT’s superiority in learning continuous space-time representations, outperforming baselines in diverse forecasting and reconstruction tasks. SCENT is the first single-step training model for learning continuous spatiotemporal representations, eliminating multi-stage optimization bottlenecks. Its scalability makes it suitable for large-scale applications in geophysics, astronomy, epidemiology, and nuclear physics (Reichstein et al., 2019; Gabbard et al., 2022; Massucci et al., 2016; Pata et al., 2024). Future work will focus on expanding SCENT’s adaptability to extreme-scale datasets and real-world deployments.

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